

Liangliang Zhang

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EDUCATION

Rensselaer Polytechnic Institute, Troy, USA

August 2024 - Present

Ph.D. in Computer Science advised by Prof. Yao Ma

Central South University, Changsha, China

September 2022 - June 2024

B.S. in Software Engineering

RESEARCH INTEREST

Large Language Models, GraphRAG, Knowledge Graphs, Graph Neural Networks, Trustworthy AI, Data-Centric AI, Data Condensation.

SELECTED HONORS AND AWARDS

- **KDD 2024 Student Travel Award** June 2024
Awarded for outstanding contributions and participation at the KDD 2024 conference.
- **Tutorial Presentation at KDD 2024** May 2024
Co-presented a tutorial on "Safe Multi-Modal Machine Learning" at the KDD 2024 conference, showcasing advancements and best practices in the field.
- **Academic Year Scholarship, School of Computer Science, Central South University**
Awarded the scholarship **four times** of academic excellence and outstanding performance.

CONFERENCE AND JOURNAL PUBLICATIONS

- [1] **Zhang, Liangliang**, Z. Jiang, H. Chi, H. Chen, M. Elkoumy, F. Wang, Q. Wu, Z. Zhou, S. Pan, S. Wang, et al. "Diagnosing and Addressing Pitfalls in KG-RAG Datasets: Toward More Reliable Benchmarking". In: *In Proceedings of the Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS)* (2025).
- [2] **Zhang, Liangliang**, H. Bao, and Y. Ma. "Extending Graph Condensation to Multi-Label Datasets: A Benchmark Study". In: *Transactions on Machine Learning Research (TMLR)* (2025).
- [3] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and J. Bu. "Learning to Reduce the Scale of Large Graphs: A Comprehensive Survey". In: *ACM Transactions on Knowledge Discovery from Data* 19.5 (2025), pp. 1–25.
- [4] Zhao, Tianyi*, **Zhang, Liangliang***, Y. Ma, and L. Cheng. "A Survey on Safe Multi-Modal Learning System". In: *In Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)* (2024). * contributed equally to this work.
- [5] **Zhang, Liangliang**, M. Chen, X. Hu, and L. Deng. "Graph Convolutional Network and Contrastive Learning Small Nucleolar RNA (snoRNA) Disease Associations (GCLSDA): Predicting snoRNA–Disease Associations via Graph Convolutional Network and Contrastive Learning". In: *International Journal of Molecular Sciences* 24.19 (2023), p. 14429.
- [6] D. Liu, X. Li, **Zhang, Liangliang**, X. Hu, J. Zhang, Z. Liu, and L. Deng. "Hgnnlda: predicting lncrna-drug sensitivity associations via a dual channel hypergraph neural network". In: *IEEE/ACM transactions on computational biology and bioinformatics* 20.6 (2023), pp. 3547–3555.

- [7] S. Wang, **Zhang, Liangliang**, and H. Fan. “An Enhanced Multi-Constraint Optimization Algorithm for Efficient Network Topology Generation”. In: *Mathematics* 11.16 (2023), p. 3456.

PREPRINTS AND SUBMISSIONS

- [8] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and B. Jiajun. “A Survey on Graph Condensation”. In: *arXiv preprint arXiv:2402.02000* (2024).

TEACHING & RESEARCH EXPERIENCE

DAMI Lab | *Research Assitant* January 2025 – Present
Research on Data-Centric AI (Data Augmentation, Active Learning, Data Valuation, Data Condensation) and Large Language Models (GraphRAG, Knowledge Graphs for LLMs).

RPI CSCI 1100: Computer Science, | *Teaching Assitant* August 2024 – December 2024
Graded exams and assignments, proctored tests, led lab sessions, and held office hours to support student learning (20 hrs/week).

Competitive Programming Club | *Team Member* July 2022 – September 2023
Research on Data Mining and Graph Neural Networks and Bioinformatics.

Big Data Computing | *Team Member* September 2021 - September 2022
Researched and implemented swarm intelligence algorithms for resource scheduling.

INTERNSHIP

IBM Yorktown Heights May - August 2025

- Developed a large-scale data product retrieval benchmark integrating structured tables and unstructured text for enterprise data discovery.
- Built a multi-vector hybrid retrieval framework unifying semantic and lexical representations via rank fusion, achieving high recall and robustness at scale.
- Established reproducible evaluation protocols for metadata-only retrieval, supporting scalable and consistent benchmarking.

Institute of Automation, Chinese Academy of Sciences March - September 2023

- Enhanced skills in Linux and Docker for system operations.
- Debugged deep learning interfaces and developed attack algorithms.
- Optimized CW attacks and implemented AutoAttack and CopyCAT methods.

Hunan Wuche Education Technology Co., Ltd. August - September 2022

- Co-designed an online tank battle game using JavaFX and FXGL.
- Led project design and presentations, enhancing Java expertise in UI, SQL, and Javax.mail.

China Hi-Tech Group June - July 2022

- Developed a shopping mall app with Vue and SpringBoot.
- Directed program design, task coordination, and team meetings.
- Gained proficiency in SpringBoot, Vue, and JavaScript.

Guangdong QST Green Soft Training Education Technology Co., Ltd. June - July 2021

- Created an online Gomoku game in C++.
- Implemented user authentication, game modes, and enhanced C++ skills.